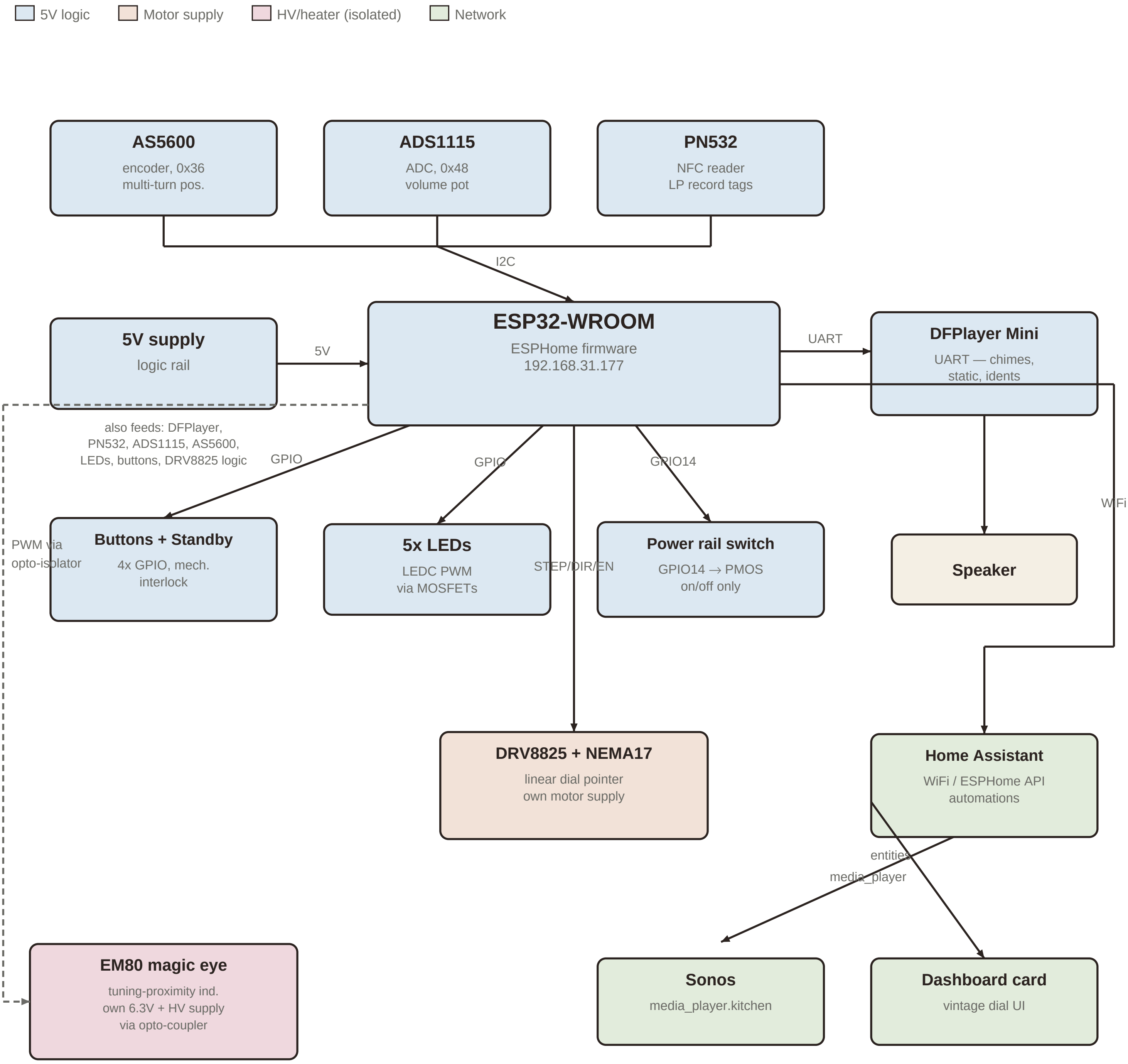
Radiola — system block diagram

ESP32 / ESPHome internet radio — motorized dial, NFC, magic eye, Home Assistant



Design notes

- Motor supply (NEMA17/DRV8825 coils) and the EM80's 6.3V/HV supply are physically separate from the shared 5V logic rail — their current transients and switching noise would otherwise disturb the I2C bus and the AS5600 encoder reading.
- AS5600 must stay on the always-on rail (not the switched power rail) so dial position is still tracked while the radio is in standby.
- EM80 control lines (heater switch, grid PWM) should pass through opto-isolation — see the reference schematic appended at the end for the 2N7000 + MAX1771 circuit.
- DRV8825 STEP/DIR/EN pins shown here (GPIO13/5/15) are substitution placeholders in radiola1.yaml — confirm against actual wiring before flashing.

Radiola — full build documentation

ESP32 / ESPHome motorized internet radio, with encoder-tracked linear dial, NFC “record” playback, DFPlayer sound effects, EM80 magic eye tuning indicator, and Home Assistant / Sonos integration.

1. Overview

Radiola is a DIY internet radio built into a vintage chassis. Tuning works like the original set: a hand-turned knob drives a linear dial pointer along a printed band, stations sit at fixed positions, and static hisses between them. Unlike the original, the ESP32 also knows the pointer's exact position at all times (via a magnetic encoder) and can drive the pointer itself with a small stepper motor — enabling soft mechanical limits, snap-to-station, seek, a standby “parking” motion, and remote tuning from Home Assistant.

2. Subsystems

ESP32-WROOM (main controller)

Runs ESPHome firmware (radiola1.yaml). Talks to all sensors and actuators locally over I2C/UART/GPIO, and to Home Assistant over WiFi via the ESPHome native API. Static IP 192.168.31.177.

AS5600 magnetic rotary encoder (I2C, 0x36)

Absolute single-turn angle sensor reading a magnet on the dial mechanism. Polled every 50 ms; wraparound between samples is accumulated into a multi-turn position (persisted to flash) so the firmware always knows exactly where the pointer is, even across power cycles and standby. Must be powered from the always-on logic rail, not any switched supply, so tracking continues in standby.

ADS1115 ADC (I2C, 0x48)

Reads the volume potentiometer on channel A1. The former dial potentiometer on A0 was retired once the AS5600 took over tuning — channel A0 is currently free.

PN532 NFC reader (I2C)

Detects tagged “records” placed on the radio. In LP mode, a recognized tag triggers album playback on the Sonos via a Home Assistant service call.

DFPlayer Mini (UART)

Plays local WAV files from a microSD card: startup/shutdown chimes, station-tuning static (volume-scaled to the knob and to distance-from-station), and seek sounds. Actual music playback happens on a Sonos speaker via Home Assistant, not through this module.

DRV8825 stepper driver + NEMA17 motor (linear dial)

Drives the physical pointer along the dial cord. EN is held low (driver disabled, knob free-spinning) during normal hand tuning; the firmware energizes it only near the calibrated soft endstops (spring-back — assists turning back in range, resists going further out, never locks the knob) or during a commanded move (seek, snap-to-station, standby park, LP progress). Powered from its own dedicated motor supply, isolated from the logic rail, since stepper current transients would otherwise disturb the 5V rail.

5x status LEDs (LEDC + MOSFET outputs)

Dimmable via PWM. Used for the startup/shutdown “wake-up” sweep animation and as calibration feedback (blinking during the limit-setting sequence).

4x latching buttons + standby contact plate (GPIO)

Mechanically interlocked piano-key buttons (pressing one releases any other). Button functions: mode select (radio/LP), seek, and the 5-press calibration trigger. A separate spring contact plate detects the physical standby position and cuts the power rail via a GPIO-driven switch (converted from an earlier dimmable-light implementation, since PWMing a power rail risked chopping the DFPlayer's supply).

EM80 magic eye tuning indicator

A vacuum “magic eye” tube (EM80/6BR5) whose shadow angle narrows as the dial approaches a station, driven from the same tuning-distance signal used for the static “swim” effect. Two ESP32 GPIOs drive it through 2N7000 MOSFETs: one switches the 6.3V heater on/off, the other PWMs the control grid (~5 kHz, RC-filtered) to vary the shadow angle. Anode/target voltage comes from a MAX1771-based boost module (5V in, adjustable output, configured here to roughly 80V rather than its typical ~200V). Full pin-out, wiring, and safety notes are reproduced from the reference schematic at the end of this document.

Home Assistant + Sonos + dashboard card

HA automations watch the ESP32's published sensors (Dial Mode, Radiola Preload Station, LP Progress) to start/stop Sonos playback, preload the next station's playlist muted while the dial is still in the static gap, and ramp volume on arrival. A custom Lovelace card renders a vintage dial face with a live needle mirroring the physical pointer, a mode badge, station/track label, and play/pause control.

3. Wiring schema

Literal ESP32 pin assignments for each subsystem, from radiola1.yaml. DRV8825 pins are substitution placeholders — confirm and update before wiring. For the EM80 magic eye's own pin-out and wire colors, see the reference schematic appended at the end.

ESP32 pin	Function	Connects to	Notes
GPIO21	I2C SDA	AS5600, ADS1115, PN532	Shared bus
GPIO22	I2C SCL	AS5600, ADS1115, PN532	Shared bus
GPIO17	UART TX	DFPlayer Mini RX	
GPIO16	UART RX	DFPlayer Mini TX	
GPIO27	LEDC PWM	LED 1 (via MOSFET)	
GPIO25	LEDC PWM	LED 2 (via MOSFET)	
GPIO32	LEDC PWM	LED 3 (via MOSFET)	
GPIO33	LEDC PWM	LED 4 (via MOSFET)	
GPIO26	LEDC PWM	LED 5 (via MOSFET)	
GPIO14	Digital out	PMOS gate (power rail switch)	On/off only — never PWM
GPIO23	Digital in, pull-up	Button 1	Mechanically interlocked
GPIO12	Digital in, pull-up	Button 2	Avoid — ESP32 boot-strapping pin
GPIO18	Digital in, pull-up	Button 3	
GPIO19	Digital in, pull-up	Button 4	
GPIO4	Digital in, pull-up	Standby contact plate	Plate on GND = active; away = standby
GPIO13*	Step pulse	DRV8825 STEP	*substitution placeholder
GPIO5*	Direction	DRV8825 DIR	*substitution placeholder
GPIO15*	Enable, active-low	DRV8825 EN	*substitution placeholder; floats high at boot = off
GPIO (TBD)	Digital/PWM out	EM80 heater switch (2N7000 Q1 gate)	Via opto-isolator recommended
GPIO (TBD)	PWM ~5kHz	EM80 grid control (2N7000 Q2 gate)	Via opto-isolator recommended

4. Power budget (5V logic rail)

Estimated current draw for everything sharing the 5V logic supply. The NEMA17 motor and the EM80's heater/HV supply are deliberately kept off this rail (see Subsystems).

Component	Typical	Peak	Notes
ESP32-WROOM	~80 mA	500 mA	WiFi TX bursts spike hard and briefly
DFPlayer Mini	40–60 mA	300–500 mA	Volume- and speaker-load dependent
PN532	~20 mA	100–150 mA	Peaks only while energizing a tag
5x LEDs	20–40 mA	75–100 mA	All five lit (startup sweep) is the peak case
AS5600	6.5 mA	~7 mA	Always on, incl. in standby
DRV8825 logic (VDD only)	4 mA	10 mA	Coil current is on the separate motor supply
ADS1115	0.15 mA	0.26 mA	Negligible
Pot / pull-ups / misc	~1–2 mA	~2 mA	Negligible
Total	**~190 mA**	**~1.2 A**	**≈ 1 W typical / ≈ 6 W peak — supply: 5V/2A min**

If the EM80 heater/HV boost shares the 5V rail, budget an additional ~580–650 mA (not recommended — give it a dedicated supply instead, both for current headroom and to keep switching noise off the I2C bus).

5. Firmware behavior summary

- **Soft endstops:** dial_min/dial_max calibrated by hand; the motor springs the pointer back toward the safe span near either limit rather than locking it in place.
- **Stations:** four positions at 20/40/60/80% of the usable dial span, each with a capture zone.
- **Snap-to-station:** once the hand-turned dial settles near a station, the motor centers it.
- **Seek:** sweeps to the next station with static hissing until arrival.
- **Static “swim”:** hiss volume scales with distance from the nearest station and the volume knob.
- **Preload:** while tuning through static, a sensor names the station being approached (direction-aware, with jitter hysteresis) so Home Assistant can start that playlist muted before arrival.
- **Standby park:** the pointer glides to the left end while the shutdown chime plays, then the power rail cuts.
- **Calibration:** 5x press of Button 4 enters calibration; Button 3 captures the left then right physical limits.
- **Failsafe:** if the encoder goes silent for more than 1 second, the motor is disabled and moves are locked out until recalibrated.

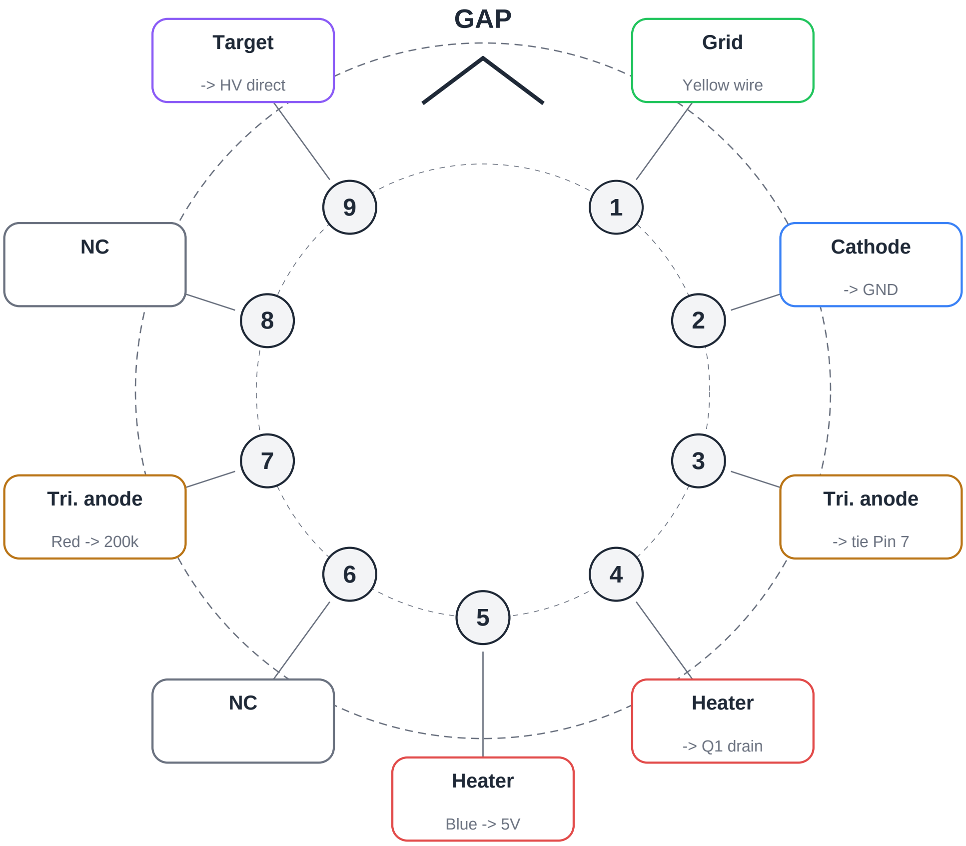
Source: esphome-devices/radiola (radiola1.yaml, README.md, homeassistant/). Values marked as estimates should be verified against real hardware. The EM80 pin-out, circuit schematic, and parts list on the following pages are reproduced from the supplied reference document.

EM80 Magic Eye Tube

ESP32 Radiola Project — Wiring Reference

Pin Layout — Bottom View (looking at pins)

Pins numbered clockwise from gap (standard Noval B9A)



Pin Reference Table

Pin	Function	Connection	Wire
1	Grid (control input)	Q2 drain + 1M to GND	Yellow
2	Cathode	GND	Black
3	Triode anode (internal)	Tie to Pin 7 externally	-
4	Heater	Q1 drain (MOSFET switch)	Black
5	Heater	5V supply direct	Blue
6	No connection	-	-
7	Triode anode	200k resistor to HV+	Red
8	No connection	-	-
9	Target anode (display)	HV+ direct (~200V)	-

How the EM80 Works

Understanding each element of the tube

Heater (Pin 4 & 5)

A thin filament inside the tube, like a light bulb element. Put voltage across these pins and the cathode heats up so it can emit electrons. Takes 10-15 seconds to warm up.

Cathode (Pin 2)

The electron source. When heated by the filament, it emits electrons toward the target. Connected directly to GND.

Target Anode (Pin 9)

The fluorescent screen that glows green. Electrons from the cathode hit this surface. Gets HV directly. The shadow appears where electrons are deflected away.

Grid (Pin 1)

The control input. A small voltage change here (0V to -14V) controls how much the triode conducts. This is where the ESP32 PWM signal controls the shadow.

Triode Anode (Pin 3 & Pin 7)

The amplifier output. Connected to HV through a 200k load resistor. The voltage here determines the shadow position on the display.

How Grid Self-Bias Works

The grid (Pin 1) is connected to GND through a large resistor (1M or more).
Inside the tube, some electrons from the hot cathode land on the grid. This tiny current flows through the resistor to ground. By Ohm's law, even a few microamps through 1M creates several volts of drop, making the grid negative relative to ground. This is called 'grid leak bias' and it happens naturally.

When Q2 (MOSFET) turns on, it shorts Pin 1 directly to GND, overriding the self-bias and forcing the grid to 0V. The PWM duty cycle controls how much time Q2 spends on vs off, effectively varying the average grid voltage.

Shadow Behavior

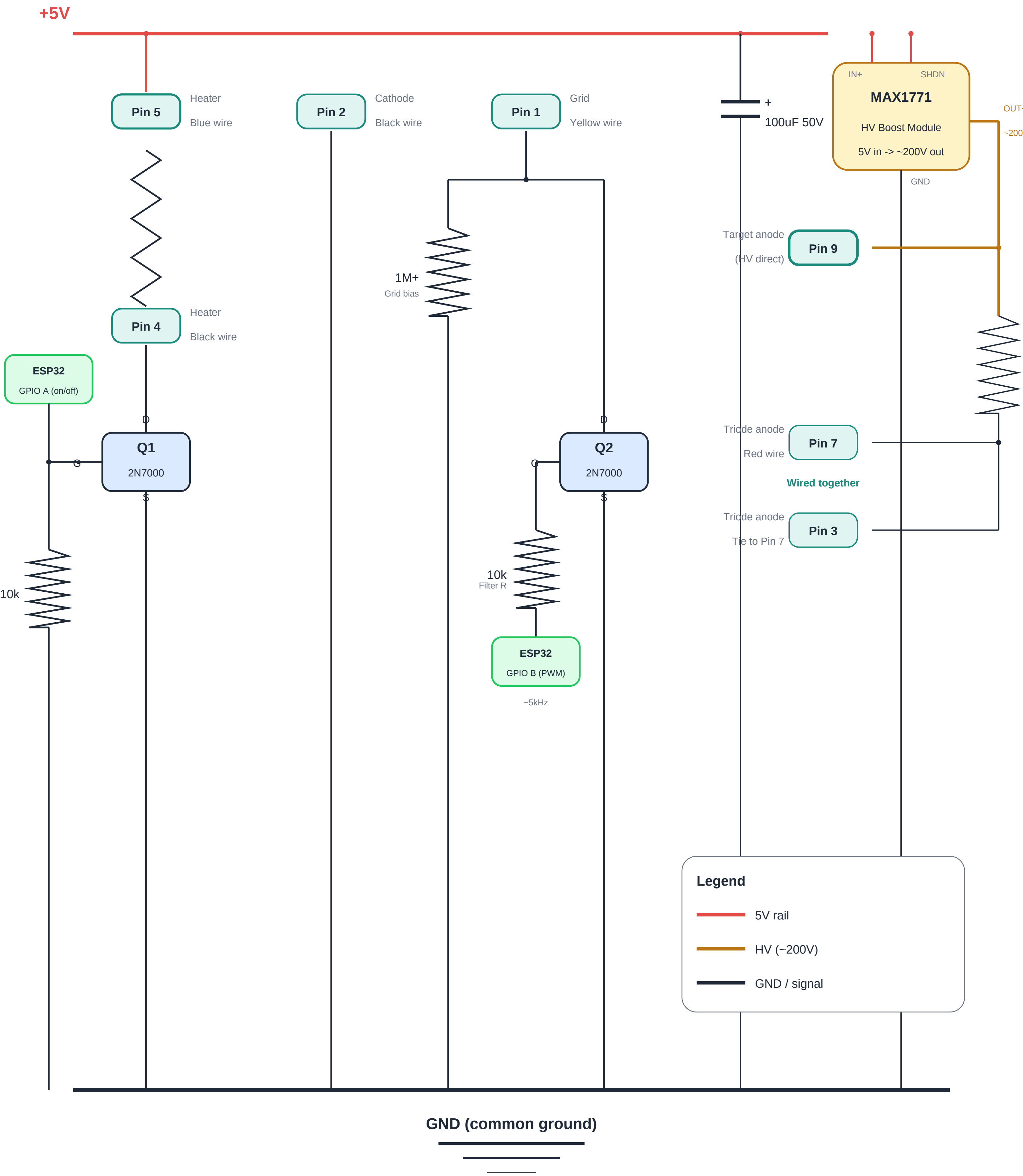
PWM low (0%)	-> Q2 off -> Grid self-biases negative -> Shadow closes -> MORE green glow = TUNED
PWM high (100%)	-> Q2 on -> Grid clamped to 0V -> Shadow opens -> LESS green glow = BETWEEN

Note on bias resistor value:

Start with 1M. If shadow range is too small, add more 1M resistors in series.
2-5M gives more range. Larger resistor = more negative bias = wider shadow sweep.

Complete Circuit Schematic

ESP32 Radiola — EM80 with heater switch and shadow control



Build Reference

Parts list, connections, and safety

Parts List

Component	Value / Type	Qty	Notes
EM80 (6BR5)	Magic eye tube	1	Noval 9-pin base
2N7000	N-ch MOSFET	2	Q1: heater, Q2: shadow
MAX1771 module	5V to 150-220V DC	1	With SHDN pin
Resistor	200k	1	Triode load resistor (HV side)
Resistor	1M (or more)	1+	Grid bias (try 2-5M for more range)
Resistor	10k	2	Q1 pull-down + Q2 filter
Capacitor	100uF 50V	1	5V rail decoupling
Noval socket	B9A 9-pin	1	For the EM80 tube

ESP32 to Breadboard (4 wires)

GPIO A	Digital output	Q1 gate (heater on/off) + HV module SHDN
GPIO B	PWM ~5kHz	Q2 gate via 10k filter (shadow control)
+5V	Power	5V rail on breadboard
GND	Ground	Common ground rail

Breadboard to Tube Socket (6 wires)

Pin 1	Grid (yellow)	Q2 drain + 1M junction on breadboard
Pin 2	Cathode (black)	GND rail
Pin 4	Heater (black)	Q1 drain on breadboard
Pin 5	Heater (blue)	5V rail
Pin 7	Triode (red)	200k to HV+ (also jumper Pin 3 to Pin 7 at socket)
Pin 9	Target	HV+ direct from module

Safety Notes — HIGH VOLTAGE

- The HV module outputs ~200V DC which can be lethal.
- NEVER touch the circuit while powered on.
- After powering off, wait at least 60 seconds before touching the HV side.
- Use a multimeter to verify 0V on HV output before working on it.
- Keep HV wiring physically separated from low voltage breadboard.
- Solder HV connections on perfboard, not breadboard.
- Insulate all exposed HV connections with heat shrink tubing.
- Use only one hand when measuring or probing the HV side.